Lab Five-Six – Data Wrangling in R

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Use the pipe operator (`|>`). It makes the data wrangling tasks much easier to write.
- Use `head(...)` or `glimpse(...)` often to ensure your code did what you wanted it to and that you haven't accidentally deleted their entire dataset with a bad filter.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!

Timeline

- Week of 2/16: Aim to complete the data wrangling sections
- Week of 2/23: Aim to complete the data summaries sections

1 Biomedical Science Data

Kitsberg et al. (2025) collect data on the Human Cytomegalovirus (HCMV). HCMV is a member of the herpesvirus family – the same family that includes: chickenpox, mononucleosis (mono), and cold sores. Prevalence is high with a worldwide seroprevalence of 24% to 100% (Gupta and Shorman 2025). The virus persists in a dormant state for the entire life of those infected. Primary cytomegalovirus infection is often asymptomatic but can present with flu-like symptoms. The infection can reactivate due to stress, aging, or a weakened immune system.

The researchers obtained blood samples from healthy donors, males and females, aged 25-45. They infected the cells of primary (human donor's blood) or THP1 (purchased cell lines) monocytes ($n = 87$ and $n = 112$, respectively) and macrophages ($n = 93$ and $n = 170$, respectively) and counted the number of viral genomes in the nuclei.

Note that the original donor of the THP-1 cell line was a 1-year-old Japanese boy. He was treated for Acute Monocytic Leukemia at Tohoku Hospital Pediatrics (THP-1; the first cell line there). Because they are malignant cancer cells, they can divide indefinitely, breaking the usual cell managment that usually reigns in rogue cells. Therefore, these cells can divide indefinitely in a lab environment. THP-1 cells, and those like them, serve as such a reliable, uniform model for medical research and improve reproducability.

Preliminary Question: Are THP-1 cells a good stand-in for human cells?

Research Question: Is the virus more prevalent in the nucleus of monocytes or macrophages, explaining why HCMV stays dormant in monocytes but goes active in macrophages?

1.1 Data Wrangling

1.1.1 Part A

Go to the website and download the source data https://www.nature.com/articles/s41467-025-68063-y#Sec30.

Describe the process – do not leave out any steps – for how you can load the data for Figures 3b and 3d from the manuscript into `R`.

Hint: You can use packages like `readxl` (Wickham and Bryan 2025) but you will find it substantially easier to open the `.xls` file using your computer's spreadsheet software and create a new `.csv` file with the data you need.

Solution After downloading the excel data from the website, I created my own csv file of just the figure 3b data. Because the data's headers were doubled with subheaders, I moved all of the THP-1 data columns under the "sample" header along with the primary data because they were individually sectioned before. This way, I had 2 very long columns with proper headings: "sample" for whether it was primary or THP-1 and "viruses within nucleus". I then read the csv file and loaded the data to an object named fig3bData with the code below.

```r
fig3bData = read.csv("Nature.comFigure3b.csv")
```

1.1.2 Part B

The data need to be cleaned. Note that the sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected) are all recorded in one column `sample`, separated with underscores. Use `separate(...)` from the `tidyverse` package to separate the `sample` column into the `SampleType`, `CellType`, `Treatment` columns. Save the object as `dat.kitsberg.cleaned`.

Hint: Use `?separate(...)` to see how it works.

Solution

```r
dat.kitsberg.cleaned = separate(
    fig3bData,
    col = "sample",
    into = c("SampleType", "CellType", "Treatment"),
  )
```

1.1.3 Part C

The researchers are only interested in observations in the treatment group (12hpi). Remove uninfected samples from the data. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

```r
dat.kitsberg.cleaned = filter(
    dat.kitsberg.cleaned,
    Treatment == "12hpi"
)
```

1.1.4 Part D

When `CellType` is "MDM", it is a Monocyte-Derived Macrophages. That is, the cell type should be macrophages which is coded as "mac". Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```r
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  mutate(CellType = if_else(CellType== "MDM", "mac", CellType))

table(dat.kitsberg.cleaned$CellType)
```

```

  mac  mono PDGFR 
  263   199    84 
```

1.2 Part E

The researchers are only interested in monocytes and macrophages. Remove any other cell types. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

```r
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  filter(CellType == "mac" | CellType == "mono")

table(dat.kitsberg.cleaned$CellType)
```

```
 mac mono
 263  199
```

1.3 Part F

Clean up the data entry. Replace the shorthand the researchers used with the neat sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected). Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```r
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  mutate(SampleType = if_else(SampleType == "primary", "Healthy Donor", SampleType))|>
  mutate(SampleType = if_else(SampleType == "THP1", "THP-1", SampleType)) |>
  mutate(CellType = if_else(CellType== "mono", "Monocytes", CellType)) |>
  mutate(CellType = if_else(CellType== "mac", "Macrophages", CellType)) |>
  mutate(Treatment = if_else(Treatment == "12hpi", "12 Hours Post-Infection", Treatment))
```

1.4 Part G

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.kitsberg.cleaned`.

Note: You will use this in the next section

Solution

```r
fig3bData = read.csv("Nature.comFigure3b.csv")

dat.kitsberg.cleaned = separate(
    fig3bData,
    col = "sample",
    into = c("SampleType", "CellType", "Treatment"),
  ) |>
  filter(
    Treatment == "12hpi"
  ) |>
  mutate(CellType = if_else(CellType== "MDM", "mac", CellType)) |>
  filter(CellType == "mac" | CellType == "mono") |>
  mutate(SampleType = if_else(SampleType == "primary", "Healthy Donor", SampleType))|>
  mutate(SampleType = if_else(SampleType == "THP1", "THP-1", SampleType)) |>
  mutate(CellType = if_else(CellType== "mono", "Monocytes", CellType)) |>
  mutate(CellType = if_else(CellType== "mac", "Macrophages", CellType)) |>
  mutate(Treatment = if_else(Treatment == "12hpi", "12 Hours Post-Infection", Treatment))
```

1.5 Data Summaries

1.5.1 Part A

Recreate Figure 3b as depicted in the paper (see https://www.nature.com/articles/s41467-025-68063-y#Fig3). You can ignore the significance comparison because we haven't done that yet.

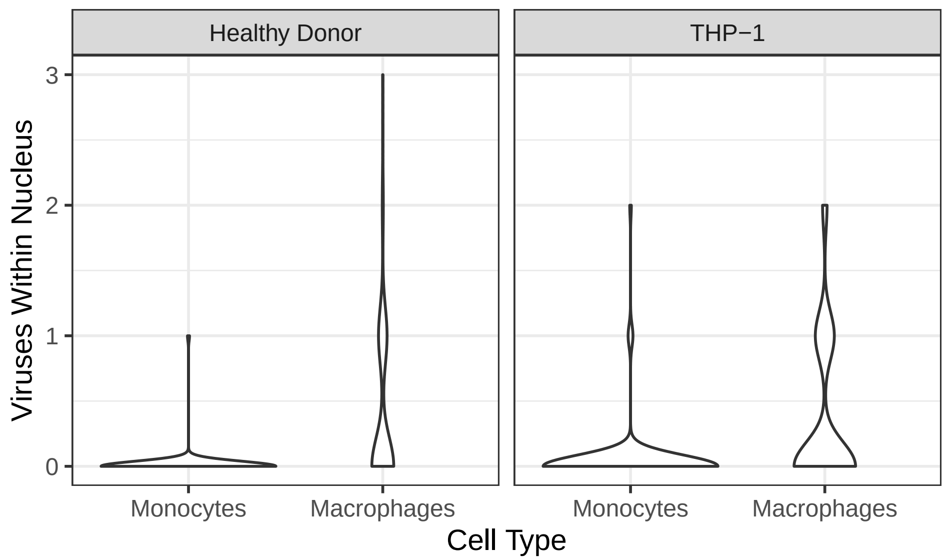
Hint: I would use `geom_violin(...)` for Figure 3b I also expect your plots to look nicer – better labels, better scaling, etc. Don't forget to use the proper formatting using the syntax in the Quarto cheatsheet.

Solution

```
dat.kitsberg.cleaned <- dat.kitsberg.cleaned |>
  mutate(CellType = factor(CellType,
                           levels=c("Monocytes", "Macrophages")))
```

```
ggplot(data = dat.kitsberg.cleaned) +
  geom_violin(aes(x = CellType, y = viruses.within.nucleus)) +
  theme_bw() +
  labs(x = "Cell Type",
       y = "Viruses Within Nucleus") +
  facet_wrap(~SampleType)
```

Figure 1: Healthy Donor vs. THP-1



1.5.2 Part B

Compute a table containing relevant numerical summaries for Figure 3b.

Hint: I would use `group_by(...)` and `summarize(...)` to complete this. Don't forget to use the proper formatting using `knitr::kable(...)` as described in the Quarto Cheatsheet.

Solution

```
celltype.summary = dat.kitsberg.cleaned |>
  group_by(CellType) |>
  summarize(
    mean = mean(viruses.within.nucleus),
    sd = sd(viruses.within.nucleus),
    median = median(viruses.within.nucleus),
    IQR = IQR(viruses.within.nucleus),
    skew = e1071::skewness(viruses.within.nucleus),
    ekurt = e1071::kurtosis(viruses.within.nucleus)
  ) |>
  knitr::kable(digits = 3)
celltype.summary
```

CellType	mean	sd	median	IQR	skew	ekurt
Monocytes	0.030	0.199	0	0	7.228	56.921

CellType	mean	sd	median	IQR	skew	ekurt
Macrophages	0.342	0.583	0	1	1.606	2.081

2 Ship Valuation

Figure 2: This is a ship-shipping ship shipping shipping ships.

Bal, Akan, and Gencer (2025) aim to build a model for predicting the value of Supramax and Ultramax ships using both current and past market data. Their goal was to build a model that only uses information anyone can easily look up.

The researchers collected 17 years of real ship sale prices (2005–2022), covering $n = 1819$ Supramax and Ultramax ship sales.

Research Question: Can we predict Supramax and Ultramax ship prices?

2.1 Data Wrangling

2.1.1 Part A

I downloaded the data from linked in the article (https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319073). I have included the data in this lab folder as `ShipValuation-Bal25.csv`. Load the data into `R`.

Solution

```
shipData = read_csv("ShipValuation-Bal25.csv")
```

2.1.2 Part B

The price of each Supramax and Ultramax ship is reported on the $\log_{10}$ scale. Create a new column called `price` that contains the price in millions of dollars.

$$\text{Price} = \frac{10^{\text{SPLog}}}{1000000}$$

Save the object as `dat.bal.cleaned`.

Hint: I would use the `mutate(...)` function.

Solution

```
dat.bal.cleaned = shipData |>
  mutate(price = (10^SPLog)/1000000)
```

2.1.3 Part C

The authors recorded the country that each Supramax or Ultramax ship was built in in the columns `Japan`, `China, People's Republic Of`, `Philippines`, `Korea, South`, `Others Country Build`. These columns are marked `1` to indicate that is the country of origin and `0` otherwise. Create a new column called `OriginCountry` that contains a clean labels for the country in which the ship was built. Update `dat.bal.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(OriginCountry = case_when(`Japan` == "1" ~ "Japan",
  `China, People's Republic Of` == "1" ~ "China, People's Republic Of",
  `Philippines` == "1" ~ "Philippines",
  `Korea, South` == "1" ~ "Korea, South",
  `Others Country Build` == "1" ~ "Others Country Build"))
```

2.1.4 Part D

The authors recorded each Supramax and Ultramax ship's main engine manufacturer in the columns `Caterpillar`, `MAN-B&W`, `Wartsila`, `Mitsubishi`. These columns are marked `1` to indicate that is the manufacturer and `0` otherwise. Create a new column called `Manufacturer` that contains a clean labels for the manufacturer. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous question.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(Manufacturer = case_when(`Caterpillar` == "1" ~ "Caterpillar",
  `MAN-B&W` == "1" ~ "MAN-B&W",
  `Wartsila` == "1" ~ "Wartsila",
  `Mitsubishi` == "1" ~ "Mitsubishi"))
```

2.1.5 Part E

The researchers recorded whether each Supramax and Ultramax ship is currently covered by a Protection and Indemnity (P&I) Club. This can be an indicator that a ship meets the minimum seaworthiness and safety requirements of the insurer.

The authors recorded P&I clubs in the columns `Japan P&I Club`, `Gard`, `West of England`, `North of England`, `UK P&I`, `Britannia P&I`, `London P&I Club`, `Skuld`, `Steamship Mutual`, `Others`, `Unknown`. These columns are marked `1` to indicate that the ship is insured by that P&I club and `0` otherwise. Create a new column called `PI.club` that contains a clean labels for the P&I club. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(PI.club = case_when(`Japan P&I Club` == "1" ~ "Japan P&I Club",
  `Gard` == "1" ~ "Gard",
  `West of England` == "1" ~ "West of England",
  `North of England` == "1" ~ "North of England",
  `UK P&I` == "1" ~ "UK P&I",
  `Britannia P&I` == "1" ~ "Britannia P&I",
  `London P&I Club` == "1" ~ "London P&I Club",
  `Skuld` == "1" ~ "Skuld",
```

```
  `Steamship Mutual` == "1" ~ "Steamship Mutual",
  `Others` == "1" ~ "Others",
  `Unknown` == "1" ~ "Unknown"))
```

2.1.6 Part F

The researchers recorded whether each Supramax and Ultramax ship is classed by a society that requires a ship to be safe, structurally sound, and built/maintained to specific engineering standards.

The authors recorded class society in the columns `Nippon Kaiji Kyokai (IACS)`, `Lloyd's Register`, `Bureau Veritas (IACS)`, `Det Norske Veritas (IACS)`, `American Bureau of Shipping (IACS)`, `Others Class`, `With Two Classes`, `Disclassed`. These columns are marked `1` to indicate that the ship is classed by that soceity and `0` otherwise. Create a new column called `Class` that contains a clean labels for the class. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(Class = case_when(`Nippon Kaiji Kyokai (IACS)` == "1" ~ "Nippon Kaiji Kyokai (IACS)",
  `Lloyd's Register` == "1" ~ "Lloyd's Register",
  `Bureau Veritas (IACS)` == "1" ~ "Bureau Veritas (IACS)",
  `Det Norske Veritas (IACS)` == "1" ~ "Det Norske Veritas (IACS)",
  `American Bureau of Shipping (IACS)` == "1"
                          ~ "American Bureau of Shipping (IACS)",
  `Others Class` == "1" ~ "Others Class",
  `With Two Classes` == "1" ~ "With Two Classes",
  `Disclassed` == "1" ~ "Disclassed"))
```

2.1.7 Part G

The researchers recorded the flag state of each Supramax and Ultramax ship. Each flag state enforces its own set of safety, environmental, and crew welfare rules and regulations. Ships under certain flag states may be more or less valuable depending on how much the flag affects the buyer's trust in the condition of the boat.

The authors recorded flag states in the columns `Panama`, `Marshall Islands`, `Liberia`, `Malta`, `Hong Kong`, `China`, `Singapore`, `Others Flag`, `With two flags`. These columns are marked `1` to indicate that the ship has that flag state and `0` otherwise. Create a new column called `FlagState` that contains a clean labels for the flag state. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(FlagState = case_when(`Panama` == "1" ~ "Panama",
  `Marshall Islands` == "1" ~ "Marshall Islands",
  `Liberia` == "1" ~ "Liberia",
  `Malta` == "1" ~ "Malta",
  `Hong Kong, China` == "1" ~ "Hong Kong, China",
  `Singapore` == "1" ~ "Singapore",
  `Others Flag` == "1" ~ "Others Flag",
  `With two flags` == "1" ~ "With two flags"))
```

2.2 Part H

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.bal.cleaned`.

Note: You will use this in the next section

Solution

```
shipData = read_csv("ShipValuation-Bal25.csv")

dat.bal.cleaned = shipData |>
  mutate(price = (10^SPLog)/1000000) |>
  mutate(OriginCountry = case_when(`Japan` == "1" ~ "Japan",
    `China, People's Republic Of` == "1" ~ "China, People's Republic Of",
    `Philippines` == "1" ~ "Philippines",
    `Korea, South` == "1" ~ "Korea, South",
    `Others Country Build` == "1" ~ "Others Country Build")) |>
  mutate(Manufacturer = case_when(`Caterpillar` == "1" ~ "Caterpillar",
    `MAN-B&W` == "1" ~ "MAN-B&W",
    `Wartsila` == "1" ~ "Wartsila",
    `Mitsubishi` == "1" ~ "Mitsubishi")) |>
  mutate(PI.club = case_when(`Japan P&I Club` == "1" ~ "Japan P&I Club",
   `Gard` == "1" ~ "Gard",
   `West of England` == "1" ~ "West of England",
   `North of England` == "1" ~ "North of England",
   `UK P&I` == "1" ~ "UK P&I",
   `Britannia P&I` == "1" ~ "Britannia P&I",
   `London P&I Club` == "1" ~ "London P&I Club",
   `Skuld` == "1" ~ "Skuld",
   `Steamship Mutual` == "1" ~ "Steamship Mutual",
   `Others` == "1" ~ "Others",
   `Unknown` == "1" ~ "Unknown")) |>
  mutate(Class = case_when(`Nippon Kaiji Kyokai (IACS)` == "1" ~
                                     "Nippon Kaiji Kyokai (IACS)",
   `Lloyd's Register` == "1" ~ "Lloyd's Register",
   `Bureau Veritas (IACS)` == "1" ~ "Bureau Veritas (IACS)",
   `Det Norske Veritas (IACS)` == "1" ~ "Det Norske Veritas (IACS)",
   `American Bureau of Shipping (IACS)` == "1"
                             ~ "American Bureau of Shipping (IACS)",
   `Others Class` == "1" ~ "Others Class",
   `With Two Classes` == "1" ~ "With Two Classes",
   `Disclassed` == "1" ~ "Disclassed")) |>
  mutate(FlagState = case_when(`Panama` == "1" ~ "Panama",
    `Marshall Islands` == "1" ~ "Marshall Islands",
   `Liberia` == "1" ~ "Liberia",
   `Malta` == "1" ~ "Malta",
   `Hong Kong, China` == "1" ~ "Hong Kong, China",
   `Singapore` == "1" ~ "Singapore",
   `Others Flag` == "1" ~ "Others Flag",
   `With two flags` == "1" ~ "With two flags"))
```

2.3 Data Summaries

First, I provide a short description of the other variables in their data. Note that the size ordering of types of ships discussed below is:

$$\text{Handysize} < \text{Supramax} < \text{Ultramax} < \text{Panamax} < \text{Capesize}$$

- `DWT`: The deadweight tonnage, which is the maximum weight each Supramax and Ultramax ship can safely carry in metric tons. This gives a measure of how much cargo, fuel, fresh water, ballast water, provisions, and crew a ship can handle.
- `Yas`: The age in years. New Supramax and Ultramax ships were recorded as having age 1.
- `Engınges RPM`: Revolutions per minute of each Supramax and Ultramax ship's main propulsion engine. Generally speaking, lower-RPM engines are more expensive because they are more fuel efficient.

- `BACI`: The Baltic Capesize Index. This describes freight rates for Capesize dry bulk ships, which reflects supply and demand. When BACI is high, Capesize ships are more profitable in shipping, which could positively affect Supramax and Ultramax ship values as demand for dry bulk commodities might be high and large shipments could be split into several Supramax and Ultramax shipments.
  Note: This reports the current BACI. The data also contain the BACI form 1, 2, and 3 months before the sale (`BACI-1`, `BACI-2`, `BACI-3`).
- `BPNI`: The Baltic Panamax Index. This describes the cost to ship dry bulk commodities (e.g., coal or grain) on Panamax class ships. When BPNI is high, we expect Panamax ship prices to go up. Supramax and Ultramax ship values tend to follow this trend as charterers switch to these smaller vessels when Panamax rates become too expensive.
  Note: This reports the current BPNI. The data also contain the BPNI form 1, 2, and 3 months before the sale (`BPNI-1`, `BPNI-2`, `BPNI-3`).
- `BSIS`: The Baltic Supramax Index. This describes freight rates for Supramax vessels, which reflects supply and demand. When BSIS is high, Supramax and Ultramax prices are usually higher because they are more profitable in shipping.
  Note: This reports the current BSIS. The data also contain the BSIS form 1, 2, and 3 months before the sale (`BSIS-1`, `BSIS-2`, `BSIS-3`).
- `BHSI`: The Baltic Handysize Index. This describes freight rates for handysize vessels. When BHSI is high, Supramax and Ultramax ship prices are usually higher because the prices move together.
  Note: This reports the current BHSI. The data also contain the BHSI form 1, 2, and 3 months before the sale (`BHSI-1`, `BHSI-2`, `BHSI-3`).
- `LIBOR`: The London InterBank Offered Rate. This describes the average reported interest rate at which banks would lend US dollars for 3 months.

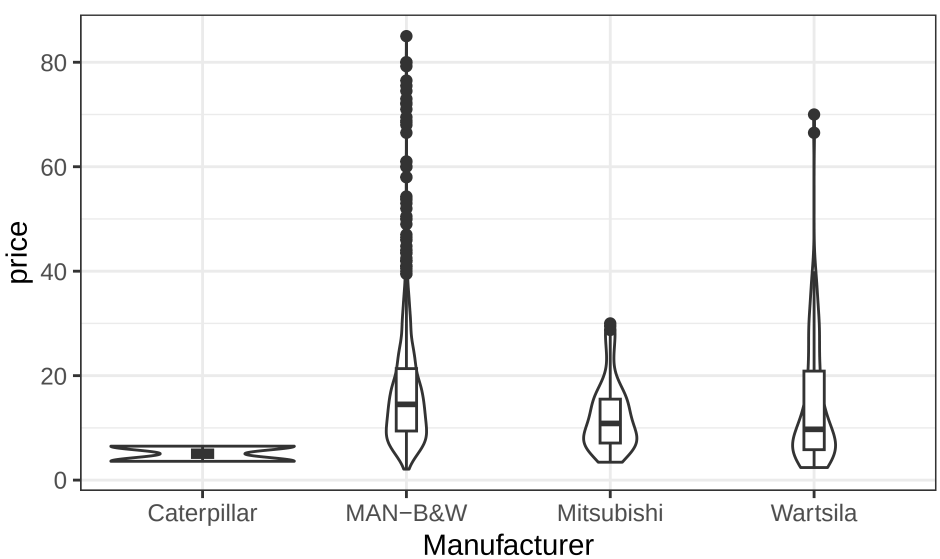
2.3.1 Part A

Choose a categorical variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary across the levels of the selected categorical variable.

Solution

```r
ggplot(data = dat.bal.cleaned |> drop_na(Manufacturer)) +
  geom_violin(aes(x = Manufacturer, y = price)) +
  geom_boxplot(aes(x = Manufacturer, y = price), width = 0.1) +
  theme_bw() +
  labs(x = "Manufacturer",
       y = "price")
```

Figure 3: Manufacturer Comparison




```r
manufacturer.summary = dat.bal.cleaned |>
  drop_na(price) |>
```

```
  group_by(Manufacturer) |>
  summarize(
    mean = mean(price),
    sd = sd(price),
    median = median(price),
    IQR = IQR(price),
    skew = e1071::skewness(price),
    ekurt = e1071::kurtosis(price)
  ) |>
  knitr::kable(digits = 3)
manufacturer.summary
```

Manufacturer	mean	sd	median	IQR	skew	ekurt
Caterpillar	5.050	2.051	5.050	1.45	0.000	-2.750
MAN-B&W	16.884	11.065	14.500	11.95	2.199	7.618
Mitsubishi	12.351	6.724	10.850	8.40	1.041	0.388
Wartsila	14.311	11.605	9.725	15.05	1.724	4.055

Manufacturer Comparison

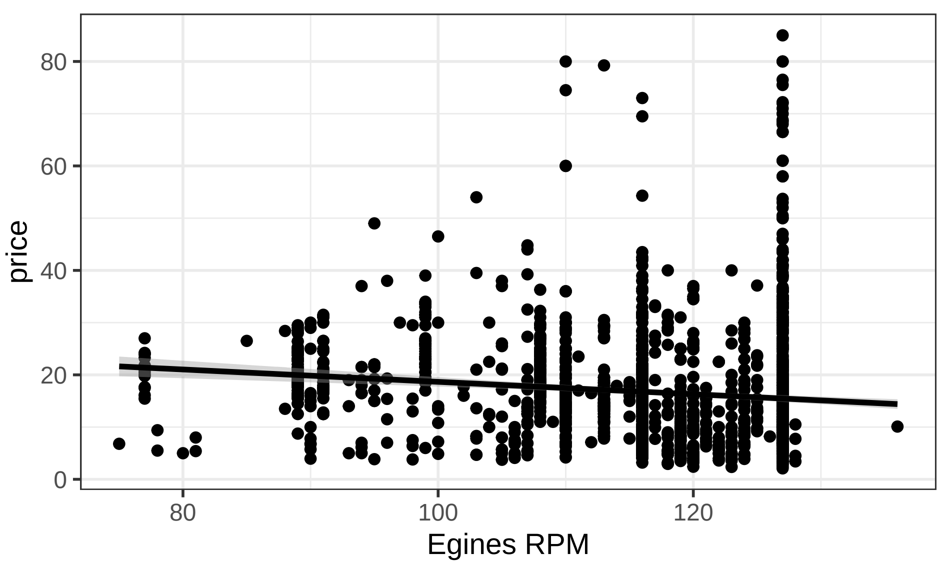
2.3.2 Part B

Choose a quantitative variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary with the selected quantitative variable.

Solution

```
ggplot(data = dat.bal.cleaned) +
  geom_point(aes(x = `Engines RPM`, y = price)) +
  geom_smooth(aes(x = `Engines RPM`, y = price), method = "lm", color = "black") +
  theme_bw() +
  labs(x = "Egines RPM",
       y = "price")
```

Figure 4: Engines RPM comparison




```
EngineRPM.summary = dat.bal.cleaned |>
  drop_na(price) |>
  group_by(`Engines RPM`) |>
  summarize(
    mean = mean(price),
    sd = sd(price),
    median = median(price),
    IQR = IQR(price),
    skew = e1071::skewness(price),
    ekurt = e1071::kurtosis(price)
  ) |>
  knitr::kable(digits = 3)
EngineRPM.summary
```

Engines RPM	mean	sd	median	IQR	skew	ekurt
75	6.800	NA	6.800	0.000	NaN	NaN
77	20.930	3.288	21.000	4.900	-0.047	-1.070
78	7.450	2.758	7.450	1.950	0.000	-2.750
80	5.000	NA	5.000	0.000	NaN	NaN
81	6.700	1.838	6.700	1.300	0.000	-2.750
85	26.500	NA	26.500	0.000	NaN	NaN
88	20.950	10.536	20.950	7.450	0.000	-2.750
89	20.329	4.042	19.500	5.000	0.067	0.143
90	14.349	8.849	14.000	9.700	0.578	-1.182
91	20.512	5.625	19.500	6.675	0.517	-0.710
93	12.667	7.095	14.000	7.000	-0.181	-2.333
94	16.785	8.987	16.500	8.500	0.589	-0.083
95	20.523	10.831	19.200	6.000	1.294	1.867
96	18.240	11.952	15.400	7.800	0.699	-1.329
97	30.000	NA	30.000	0.000	NaN	NaN
98	12.605	9.338	10.250	8.176	0.775	-1.033
99	25.043	5.881	24.500	5.000	-0.284	1.279
100	17.527	13.904	13.425	8.100	1.053	-0.429
102	16.875	1.237	16.875	0.875	0.000	-2.750
103	17.978	17.445	8.300	13.200	1.027	-0.613
104	17.450	8.515	12.500	10.250	0.463	-1.864
105	18.362	11.906	19.100	18.200	0.296	-1.374
106	7.540	2.887	7.000	1.605	1.223	1.061
107	17.831	11.653	13.850	8.650	1.099	0.100
108	21.297	4.835	21.000	6.000	0.401	0.177
109	11.000	NA	11.000	0.000	NaN	NaN
110	19.589	13.487	16.000	13.150	2.352	6.910
111	20.250	4.596	20.250	3.250	0.000	-2.750
112	11.800	6.647	11.800	4.700	0.000	-2.750
113	16.497	8.414	14.500	4.500	4.701	30.852
114	17.900	NA	17.900	0.000	NaN	NaN
115	15.017	3.898	16.000	4.360	-0.749	-1.086
116	17.474	11.528	15.400	13.000	1.789	4.867
117	19.836	9.442	19.000	15.400	0.154	-1.736
118	13.736	11.091	8.700	15.625	0.894	-0.687
119	11.449	6.801	9.300	9.012	1.035	0.148
120	13.802	9.870	11.650	14.300	0.908	-0.387
121	11.605	3.504	11.650	5.963	0.023	-1.472
122	8.598	5.887	6.500	3.027	1.545	0.994
123	11.584	9.386	7.757	10.191	1.408	1.373
124	13.414	7.806	10.800	11.200	0.796	-0.736
125	16.453	6.854	13.700	8.075	1.382	1.709
126	8.200	NA	8.200	0.000	NaN	NaN
127	16.066	12.392	11.925	10.200	2.416	7.286
128	6.544	3.215	6.125	4.206	0.187	-2.160
136	10.100	NA	10.100	0.000	NaN	NaN
NA	8.703	7.161	6.000	2.121	2.734	7.495

Engines RPM comparison

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